

Swayam Purwar

Full-Stack Developer | Real-Time Systems & Scalable Apps

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PROFESSIONAL SUMMARY

Full-Stack Developer specializing in low-latency, event-driven systems using the MERN stack. Proven track record of improving performance (up to 60%), reducing costs, and building scalable real-time applications. Focused on system design, optimization, and production-grade engineering.

TECHNICAL SKILLS

Languages	JavaScript (ES6+)	SQL	Databases	MongoDB	Redis	
Frontend	React.js	Next.js	HTML/CSS	Systems	WebSockets	Caching
Backend	Node.js	Express.js	Tools	Git	GitHub	Vercel

PROJECTS

RepoSage Prime [Live ↗](#) [GitHub ↗](#) [Next.js, LangChain, OpenAI, Pinecone](#)

- Architected a RAG-based SaaS platform enabling semantic querying across multi-document datasets with high precision.
- Optimized vector search accuracy by implementing recursive character splitting and metadata-aware chunking strategies.
- Reduced perceived user latency by 60% through the implementation of Vercel AI SDK for real-time token streaming.
- Lowered LLM API consumption costs by 30% by implementing a Redis-based semantic caching layer for frequent queries.

Kite Trading Application [Live ↗](#) [GitHub ↗](#) [React, Node.js, WebSockets, MongoDB](#)

- Engineered a high-throughput event-driven dashboard capable of handling 1,000+ concurrent market data updates per second.
- Reduced React re-renders by 35% using Memoization and specific state-slicing techniques to ensure 60FPS UI performance.
- Improved end-to-end data latency from 1.2s to ~400ms by implementing message batching and binary WebSocket protocols.
- Designed a scalable backend micro-service architecture to process and broadcast real-time price feeds via Socket.io.

Instagram Clone [Live ↗](#) [GitHub ↗](#) [React, Node.js, Express, Socket.IO, MongoDB](#)

- Implemented a robust security layer using JWT authentication and custom middleware for protected route management.
- Engineered a highly optimized MongoDB schema with compound indexing to support rapid social graph interactions.
- Developed a seamless infinite-scroll feed utilizing Intersection Observer API and backend pagination for fluid UX.
- Minimized network overhead and reduced payload size by 28% through targeted MongoDB projection and query tuning.

EXPERIENCE

Software Development Intern | Bluestock Fintech Oct 2025 – Nov 2025

- Architected and deployed robust RESTful backend APIs for financial analytics systems, ensuring 99.9% uptime for data reporting.
- Optimized database performance by 22% through the refactoring of complex Mongoose aggregation pipelines.
- Collaborated with the core engineering team to identify and resolve critical production bottlenecks while enhancing global error handling.
- Implemented standardized API documentation and integrated automated testing scripts to streamline the internal development workflow.

Freelance Full-Stack Developer | Self-Employed Sep 2024 – Present

- Engineered end-to-end web solutions for diverse clients, delivering high-performance brand websites and e-commerce MVPs.
- Consistently achieved Google Lighthouse scores of 95+ by implementing advanced image optimization and code-splitting techniques.
- Managed the complete SDLC, from initial UI/UX wireframing in Figma to final production deployment on Vercel and AWS environments.
- Developed reusable component libraries that reduced project turnaround time by 30% for subsequent client engagements.

EDUCATION

Bachelor of Computer Applications (BCA) Expected 2026 | LNCT University, Bhopal